

Formal languages and automata (COMS3003A)

Midterm test

29 September 2023

Total number of points: 35

Time: 60 mins

- ✓ 1. (10 points = 5×2 points) Prove whether the following languages are regular (i.e., state whether a language is regular and prove your claim):
 - (a) $\{01\}^* \cdot \{101\} \cdot \{01\}^*$;
 - (b) $\{ww \mid w \in \{0,1\}^*\}$.
- ✓ 2. (5 points) Suppose a language L is decidable. How many Turing machines decide L ? Justify your claim.
3. (10 points) Prove, using diagonalization, that the non self-acceptance problem for Turing machines is undecidable. In other words, prove that the language $\{\langle M \rangle \mid M(\langle M \rangle) \neq 1\}$ is undecidable.
4. (10 points) Prove that the problem of determining whether a Turing machine ever writes the symbol $\dagger$ over $\sqcup$ during the course of its computation on any input string is undecidable. In other words, prove that the language

$$L = \{\langle M \rangle \mid \exists x \text{ such that, when } M \text{ runs on } x, \text{ it writes } \dagger \text{ over } \sqcup\}$$

is undecidable.

You may assume that the following problems are undecidable:

- self-acceptance for Turing machines ("does M accept $\langle M \rangle$?");
- acceptance for Turing machines ("does M accept x ?");
- halting for Turing machines ("does M halt on x ?").

$$x = \alpha$$
$$y = \beta$$

$$\alpha > 0$$

$$\beta \geq 1$$

$$\alpha + \beta \leq p$$

$$s = p$$

$$x \geq 0$$
$$y \geq 1$$
$$x + y \leq p$$

$$z = p - \alpha - \beta$$

$$xy(p - x - y)$$
$$xyip - x^2yi - xy^2i$$